Ece Yüksel, Ph.D.

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I study how humans represent, remember, and navigate space across real and virtual environments. My research integrates behavioral, qualitative, and virtual-reality methods to understand how age, technology, and environmental scale shape spatial cognition. I use immersive VR, mixed-methods design, and advanced analytics to bridge cognitive science and real-world applications.

education

Ph.D. in Psychology *August 2026*

SCANN Lab, Department of Psychology, University of Florida, USA
Program: Behavioral and Cognitive Neuroscience
Thesis: "*Age-Related Differences in Spatial Navigation Across Real and Virtual Worlds: A Mixed-Methods Approach*"
Mentor: Steven M. Weisberg, PhD

M.S. in Psychology *May 2024*

Program: Behavioral and Cognitive Neuroscience
Thesis: *"Navigating the Metaverse: The Relation between Scale and Expertise in Spatial Knowledge of Immersive and Desktop Virtual Reality"* (Read here)
Mentor: Steven M. Weisberg, PhD

Bachelor of Science in Psychology & Architecture *2020*

TOBB University of Economics and Technology, Ankara, Turkey

skills

Quantitative Research & Data Analysis
Frequentist and Bayesian approaches
R/R Studio; analysis and visualization in JASP

Qualitative Research
Semi-structured interviewing; transcription and thematic coding; reflexive thematic analysis; mixed-methods analysis.

Virtual Reality & Experimental Programming
Unity (environment modeling, task design, scripting in C#); HTC Vive; Virtuix Omni omnidirectional treadmill; SteamVR integration; virtual navigation task development.

Data Collection & Experiment Design
VR-based behavioral experiment implementation; spatial navigation task design; mixed-methods study design and implementation.

Tools
Qualtrics; Google Forms; PsychoPy; Photoshop

Languages
English (fluent), Turkish (native), Italian (beginner)

research experience

Research Scientist *2021 - 2026*

Advisor: Steven M. Weisberg, Ph.D., Spatial Cognition and Navigational Neuroscience Lab
University of Florida, Gainesville, FL

- Developed high-quality visualizations of behavioral data using R and JASP for clear interpretation and insights from over 200 human participants
- Collected and analyzed quantitative data to inform UX decisions, optimizing user satisfaction and system interaction
- Led three independent VR-based and three research projects from concept to execution, focusing on user interaction and navigation experiences
- Presented findings on user navigation behavior and VR interaction at three international conferences

Research Intern *2020 - 2022*
PI: Didem Kadihasanoglu, Ph.D., Perception & Action Dynamics Lab
TOBB University of Economics and Technology, Ankara, Turkey

- Performed in-depth analysis of behavioral data using JASP, contributing to research that bridged real-world and virtual reality user behaviors
- Designed experimental studies on *affordances* based on comprehensive literature reviews
- Created VR environments to test hypotheses, employing both head-mounted displays and mobile devices to explore human perception and behavior in different contexts
- Presented real-world and VR findings at an international conference; published the abstract in the journal *Perception*

Research Intern *2019*
PI: Başak Şahin-Acar, Ph.D., Child & Adolescent Development Laboratory
Middle East Technical University (METU), Ankara, Turkey

Research Assistant *2019*
PI: Münire Özlem Çevik, Ph.D.
Behavioral Neurobiology Laboratory
TOBB University of Economics and Technology, Ankara, Turkey

teaching experience

Graduate Instructor, University of Florida, Gainesville, FL *2025 - 2026*
General Psychology – PSY 2012 (3 credits):

- Instructor of record for three consecutive semesters of General Psychology.
- Designed and delivered lectures, assessments, course materials, and student support.
- Demonstrated consistent improvement in student teaching evaluations across successive semesters.

UF GatorEvals – University Core Instructor Evaluation means:
Fall 2025: 3.56 | Spring 2026: 4.38 | Summer B 2026: 4.45/5.00

Graduate Teaching Assistant, University of Florida, Gainesville, FL *2021 - 2025*

- Mentored over 50 students, fostering critical thinking and problem-solving through personalized support and guidance
- Presented complex topics clearly and engagingly to diverse audiences
- Led 3 guest lectures and facilitated discussions
- Public speaking and presentation skills

Mentorship & Supervision

- Supervised 9 undergraduate research assistants in experimental design, data collection, data analysis in JASP, interview transcription, and qualitative coding.
- Coordinated and trained 6 teaching assistants for PSY2012, ensuring consistency in grading and student communication.

Undergraduate students supervised: Merrill Garlington, Karina LaRubbio, Kyle Wang, Hailey Love, Lachyn Almazova, Bailey Reeves, Liliana Roy, Lauryn Strom, Kalista Sayaf, Jaden Denney, Samuel Ginsberg

awards & scholarships

Graduate Conference Award ($750), Psychonomic Society Annual Meeting, Denver, CO *2025*
Graduate Student Travel Award ($3600), Department of Psychology, UF *2022 - 2026*
The Balkhi Experiential Learning Award ($500), Department of Psychology, UF *2025*
The Jacquelin Goldman Spring Scholarship ($14000), Department of Psychology, UF *2024 - 2025*
Departmental Nomination for Teaching Assistant Award, Department of Psychology, UF *2024*
Foundation Scholarship ($3000), Department of Psychology, UF *2021 - 2024*
Graduate Conference Award ($750), Psychonomic Society Annual Meeting in San Francisco, CA *2023*
Full Scholarship for ANFA - Moving Boundaries Course ($1800), in Venice & Umbria, Italy *2023*
Certificate of Outstanding Merit, International Student Achievement Awards, UF *2022*
Merit-Based Scholarship, TOBB University of Economics and Technology, Ankara, Turkey *2015 - 2020*

publications

1. **Yüksel, E.**, & Weisberg, S. M. (in preparation). *Navigating the metaverse: The relation between scale and expertise in spatial knowledge of immersive and desktop virtual reality.*
2. **Yüksel, E.**, & Weisberg, S. M. (in preparation). *Age-related differences in spatial navigation across real and virtual worlds: a mixed-methods approach.*
3. **Yüksel, E.,** Boogaart, Z. & Weisberg, S.M. (2025) This is not the way: global directional cues do not improve spatial learning in an immersive virtual environment. *Cogn. Research 10*, 48. DOI
4. Zheng L., Boogaart Z., McAvan A., Garren J., Doner S., Wilkes B. J., Groves W., **Yüksel E.**, Cherep L., Ekstrom A.D., Weisberg S.M. (2025) Newly trained navigation and verbal memory skills elicit changes in task-related networks but not brain structure *eLife 14*:RP106873 DOI
5. Weisberg, S. M., Barnas, A. J., Abid, H., Kumar, S., Sahoo, A., & **Yüksel, E.** (2025). *Drawing down: The structure of spatial direction representations in drawing and categorization.* Manuscript under revision, *Neuropsychologia.*
6. **Yüksel, E.** & Kadihasanoglu, D. (2021, August). *Affordance for sitting: Perception of preferred and maximum sitting heights in a virtual reality environment.* Abstract presented at the 43rd European Conference on Visual Perception (ECVP), *Perception*, 50(Suppl. 1), 87. DOI

(In preparation and under review manuscripts available upon request.)

science communication & outreach

Yüksel, E., & Weisberg, S.M. (2025, July–August). *Beyond one-size-fits-all: Tailoring navigation support for varied people and varied environments*. *Navigation News*, Royal Institute of Navigation.

conference presentations

Yüksel, E., & Weisberg, S. M. (2026, May). *Age-related differences in spatial navigation across real and virtual worlds: A mixed-methods approach.* Poster presented at CRaNECon26, Atlanta, GA, United States.

Yüksel, E., & Weisberg, S. M. (2025, October). *Age differences in spatial navigation across real and virtual worlds: A mixed-methods approach.* Poster presented at the 66th Annual Meeting of the Psychonomic Society, Denver CO, United States.

Yüksel, E., & Weisberg, S. M. (2025, September). *Age differences in spatial navigation across real and virtual worlds: A mixed-methods approach.* Poster presented at the ANFA Conference (virtual).

Yüksel, E., & Weisberg, S. M. (2025, May). *Age differences in spatial navigation across real and virtual worlds: A mixed-methods approach.* Poster presented at CRaNECon25, Atlanta, GA, United States.

Yüksel, E., & Weisberg, S. M. (2024, November). *Navigating the metaverse: The relation between scale and expertise in spatial knowledge of immersive and desktop virtual reality.* Poster presented at the 65th Annual Meeting of the Psychonomic Society, New York City, NY, United States.

Yüksel, E., Boogaart, Z., & Weisberg, S. M. (2023, November). *This is not the way: Global directional cues do not improve spatial navigation in an immersive virtual environment.* Poster presented at the 64th Annual Meeting of the Psychonomic Society, San Francisco, CA, United States.

Yüksel, E., & Weisberg, S. M. (2023, November). *Navigating the metaverse: The relation between scale and expertise in spatial knowledge of immersive and desktop virtual reality.* Poster presented at the Symposium for Individual Differences in Cognition, San Francisco, CA, United States.

Yüksel, E., Boogaart, Z., & Weisberg, S. M. (2023, February). *This is not the way: Global directional cues do not improve spatial navigation in an immersive virtual environment.* Poster presented at the 13th Annual North Central Florida Society for Neuroscience Chapter Conference, Gainesville, FL, United States.

Yüksel, E., Boogaart, Z., & Weisberg, S. M. (2022, June). *This is not the way: A global directional cue does not improve spatial navigation in an immersive virtual environment.* Talk presented at the 4th Interdisciplinary Navigation Symposium (virtual).

Yüksel, E., & Kadihasanoglu, D. (2021, August). *Affordance for sitting: Perception of preferred and maximum sitting heights in a virtual reality environment. Perception*, 50(Suppl. 1), 87. [Abstract presented at the European Conference on Visual Perception (virtual)]. DOI

professional memberships

Society for Neuroscience, American Psychological Association, Psychonomic Society, ANFA

service

Invited Reviewer, *Neuropsychological Rehabilitation* *2026*

Invited Reviewer, *32nd ACM Symposium on Virtual Reality Software and Technology (VRST) Conference Journal Track* *2026*

Invited Reviewer, *25th IEEE International Symposium on Mixed and Augmented Reality (ISMAR), Conference Journal Track* *2026*

Mentored Journal Reviewer, *Cognitive Research: Principles and Implications* *2022*

Secretary, Diversity Awareness and Affirmation Committee (DAAC), University of Florida *2023 - 2024*

Communications Team Member, Graduate Assistants United (GAU), University of Florida *2022 - 2024*

Social Media Coordinator, Turkish Student Association (TSA), University of Florida *2021 - 2022*

Editor, Journal of European Psychology Students (JEPS) *2020 - 2022*